

Disproof of conjecture 00000003524: ω_1 has two disjoint stationary subsets

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1 The conjecture and the clause refuted

Statement (English, verbatim). Definition: Subsets of ω_1 : the combinatorics of uncountable ordinals. Conjecture: Club sets and stationary sets: the intersection of stationary sets is stationary, with the reflection of stationarity bounded below closure. (stationary intersection reflection lower bound closure)

Statement (Chinese). The Chinese text says, word for word: “Definition: subsets of ω_1 : the combinatorics of uncountable ordinals. Conjecture: club sets and stationary sets: the intersection of stationary sets is stationary (*pingwen ji de jiaoji wei pingwen*) and (*qie*) the reflection of stationarity is the lower-bound closure.”

Reading. Both languages literally assert

$$(A) \quad S, T \subseteq \omega_1 \text{ stationary} \implies S \cap T \text{ stationary},$$

joined to a second clause (B), “the reflection of stationarity bounded below closure”. The Chinese text joins (A) and (B) with “and” (*qie*), so the conjecture is the conjunction $(A) \wedge (B)$. Clause (B) does not state a definite proposition: stationary reflection is a standard notion, but “bounded below closure” (“is the lower-bound closure”) has no standard meaning. We do not interpret (B). We refute (A), and we prove in Lean that $(A) \wedge Q$ is false for *every* proposition Q , so the conjunction fails however (B) is read.

We refute (A) in three forms: for two sets, for every nonempty family of stationary sets, and for the family of all stationary sets. We do *not* refute the different statement “a stationary set intersected with a club set is stationary”. That statement is true, and we prove it in Lean as a remark (Remark 7). The text says “the intersection of stationary sets”, not “of a stationary set and a club set”.

2 Definitions (as formalized)

We use Mathlib’s own definitions from `Mathlib/SetTheory/Cardinal/Cofinality/Club.lean`, unchanged.

Definition 1. ω_1 is Mathlib’s `Ordinal.omega 1`, the first uncountable ordinal. Subsets of ω_1 are subsets of the type $(\omega_1).ToType$, the well-order of order type ω_1 whose elements correspond to the countable ordinals.

Definition 2 (club). In a linear order α , a set C is a *club* (`IsClub C`) if (i) it is closed: whenever $d \subseteq C$ is nonempty and a is a least upper bound of d , then $a \in C$; and (ii) it is unbounded (cofinal):

for every x there is $y \in C$ with $x \leq y$. Mathlib states (i) as `DirSupClosed C`, which also asks that d be directed. Every subset of a linear order is directed, so this is the same condition. The Lean theorem `isClub_iff_closed_unbounded` proves the equivalence of Mathlib's `IsClub` with (i) $\wedge$ (ii). In ω_1 a set with a least upper bound is bounded, so (i) says that C contains the supremum of each of its nonempty bounded subsets. This is the usual notion of a closed subset of ω_1 .

Definition 3 (stationary). $S \subseteq \alpha$ is *stationary* (`IsStationary S`) if $S \cap C \neq \emptyset$ for every club C . The Lean theorem `isStationary_iff_meets_every_club` states this unfolding. In particular $\emptyset$ is not stationary (Mathlib's `not_isStationary_empty`), because α itself is a club.

3 Proof

Let α be an uncountable linear order with a well-founded $<$, in which every initial segment $\{x : x < a\}$ is countable and whose cofinality `Order.cof` α is not $\aleph_0$. All three hypotheses hold for $\alpha = \omega_1$: $|\omega_1| = \aleph_1 > \aleph_0$; each initial segment has cardinality $< \aleph_1$, hence is countable (Mathlib's `mk_Iio_lt`); and `cof` $\omega_1 = \aleph_1$ (Mathlib's `Ordinal.cof_toType`, `Cardinal.cof_omega_one`).

Lemma 4 (`isStationary_Ioi`). *For every $b \in \alpha$ the final segment $(b, \rightarrow) = \{a : b < a\}$ is stationary.*

Proof. If its complement $\{x : x \leq b\}$ were cofinal, every x would satisfy $x \leq b$, so $\alpha = \{x : x < b\} \cup \{b\}$ would be countable. Hence the complement is not cofinal, and a set whose complement is not cofinal is stationary (Mathlib's `IsStationary.of_not_isCofinal_compl`). $\square$

Theorem 5 (Ulam matrix; `exists_disjoint_stationary`). *α contains stationary sets S, T with $S \cap T = \emptyset$.*

Proof. For each $a \in \alpha$ the initial segment below a is countable, so we can choose $f_a : \alpha \rightarrow \mathbb{N}$ injective on $\{x : x < a\}$. For $n \in \mathbb{N}$ and $b \in \alpha$ put

$$A(n, b) = \{a \in \alpha : b < a \text{ and } f_a(b) = n\}.$$

For fixed b , $\bigcup_{n \in \mathbb{N}} A(n, b) = (b, \rightarrow)$, which is stationary by Lemma 4. Since `cof` $\alpha \neq \aleph_0$, a countable union of sets is stationary if and only if one of the sets is (Mathlib's `isStationary_iUnion_iff_of_countable`). So there is $N(b) \in \mathbb{N}$ with $A(N(b), b)$ stationary. The map $N : \alpha \rightarrow \mathbb{N}$ is not injective, because α is uncountable. So there are $b \neq b'$ with $N(b) = N(b') = n$. Put $S = A(n, b)$ and $T = A(n, b')$. If $a \in S \cap T$, then $b, b' < a$ and $f_a(b) = n = f_a(b')$. Since f_a is injective on $\{x : x < a\}$, this gives $b = b'$, a contradiction. Hence $S \cap T = \emptyset$. $\square$

Theorem 6 (`stationaryInterClaim_false`, `sInter_claim_false`, `not_isStationary_sInter_all`, `conjecture3524_false`). *In ω_1 : (1) there are stationary S, T with $S \cap T$ not stationary, so (A) is false; (2) it is false that every nonempty family of stationary sets has a stationary intersection; (3) the intersection of all stationary subsets of ω_1 is not stationary; (4) for every proposition Q , $\neg((A) \wedge Q)$.*

Proof. Take S, T from Theorem 5 with $\alpha = \omega_1$. Then $S \cap T = \emptyset$, and $\emptyset$ is not stationary. This gives (1). For (2) apply the hypothesis to the family $\{S, T\}$, and for (3) note that the intersection of all stationary sets lies inside $S \cap T$, so it is empty. (4) follows from (1). $\square$

Remark 7 (`isStationary_inter_isClub`; not part of the refutation). If $S \subseteq \omega_1$ is stationary and $C \subseteq \omega_1$ is a club, then $S \cap C$ is stationary: for any club D , $C \cap D$ is a club (Mathlib's `IsClub.inter`), so $(S \cap C) \cap D = S \cap (C \cap D) \neq \emptyset$. This is a different statement from (A).

4 Lean formalization and axiom audit

The file `lean/Conjecture3524/Basic.lean` (175 lines, import `Mathlib` only, namespace `C3524`, Lean v4.33.1, `Mathlib` v4.33.1) contains:

- `isClub_iff_closed_unbounded`, `isStationary_iff_meets_every_club`: the unfoldings of `Mathlib`’s definitions given in Section 2.
- `isStationary_Ioi`, `exists_disjoint_stationary`: Lemma 4 and Theorem 5 for a general linear order α .
- `countable_Iio_omega_one`, `cof_omega_one_toType`, `uncountable_omega_one_toType`: the hypotheses for ω_1 .
- `exists_disjoint_stationary_omega_one`: $\exists S T : \text{Set } (\omega_1).\text{ToType}, \text{IsStationary } S \wedge \text{IsStationary } T \wedge \text{Disjoint } S T$.
- `StationaryInterClaim` (the literal clause (A)): $\forall S T : \text{Set } (\omega_1).\text{ToType}, \text{IsStationary } S \rightarrow \text{IsStationary } T \rightarrow \text{IsStationary } (S \cap T)$, and `stationaryInterClaim_false` : $\neg \text{StationaryInterClaim}$.
- `sInter_claim_false`, `not_isStationary_sInter_all`: forms (2) and (3).
- `conjecture3524_false` ($Q : \text{Prop}$) : $\neg (\text{StationaryInterClaim} \wedge Q)$: the main theorem.
- `isStationary_inter_isClub`: Remark 7.

Conventions. The statements are universe-polymorphic: ω_1 is taken in `Ordinal.{u}` for an arbitrary universe u . Choice is used (via `Classical.choice`) to pick the injections f_a and the indices $N(b)$.

Axiom audit. `#print` axioms for `conjecture3524_false`, `stationaryInterClaim_false`, `sInter_claim_false`, `not_isStationary_sInter_all`, `exists_disjoint_stationary_omega_one` and `isStationary_inter_isClub` reports `[proptext, Classical.choice, Quot.sound]`. The file elaborates with no errors and no warnings. It contains no `sorry`, no `native_decide` and no added axioms.

Sources

- Wikipedia, “Stationary set”, https://en.wikipedia.org/wiki/Stationary_set: “a stationary set is a set that is not too small in the sense that it intersects all club sets”; for κ of countable cofinality “any two stationary subsets of κ have stationary intersection. This is no longer the case if the cofinality of κ is uncountable.” A stationary subset of a regular κ “can be partitioned into κ many disjoint stationary sets. This result is due to Solovay. If κ is a successor cardinal, this result is due to Ulam and is easily shown by means of what is called an Ulam matrix.”
- Wikipedia, “Club set”, https://en.wikipedia.org/wiki/Club_set: “If a set is both closed and unbounded, then it is a club set”.
- `Mathlib`, `Mathlib/SetTheory/Cardinal/Cofinality/Club.lean` (definitions `IsClub`, `IsStationary`; lemmas `IsClub.inter`, `IsStationary.of_not_isCofinal_compl`, `isStationary_iUnion_iff_of_countable`, `not_isStationary_empty`).